

Data Collection and Preprocessing Phase

Team ID	MCA
Project Title	Smartphone Usage, Digital Addiction & Well-Being Analytic
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

This phase identifies the structure and quality of the collected smartphone usage dataset. The data is explored, cleaned, transformed, and prepared for analysis in Power BI. The objective is to ensure that the data is accurate, consistent, and suitable for creating meaningful visualizations and insights.

Section	Description
Data Overview	The dataset contains information related to smartphone usage, including screen time, application usage, social media usage, notifications, sleep duration, physical activity, stress, productivity, demographic characteristics, digital addiction indicators, and well-being measures. The dataset will be examined to understand its rows, columns, records, and key attributes.
Data Cleaning	Identify and handle missing values, duplicate records, incorrect entries, inconsistent values, and invalid data.
Data Transformation	Power Query will be used to filter and sort records, remove unnecessary columns, standardize values, create calculated columns, and prepare the dataset for analysis.
Data Type Conversion	Data types will be checked and corrected where necessary. For example, age, screen time, sleep duration, physical activity, and stress scores will be converted to appropriate numerical data types, while demographic and usage categories will be stored as categorical/text values.
Column Splitting and Merging	Columns will be split or merged where required to make the data easier to analyze. For example, combined categories can

	be separated into individual fields, or related fields can be combined when appropriate.
Data Modeling	The processed data will be organized into appropriate tables and relationships. Required calculated measures and KPIs will be created for analyzing screen time, digital addiction, user groups, sleep, and well-being.
Save Processed Data	The cleaned and transformed dataset will be saved in a suitable format such as CSV or Excel and loaded into Power BI for further analysis and dashboard development.